Linear simulations with the tuned suni2D and uniLasso settings

Mean (standard error) across replicates. _std: suni2D with standardize = TRUE; _loo: uniLasso with LOO.

Model 1

Huang, AR(1) rho = 0.5; n = 100, p = 200, 15 nonzero; sigma = 1.5

	sse	rpe	support	sign	tpr	fpr
suni2D_zeta_tau	1.3577 (0.0344)	0.4378 (0.0114)	23.2150 (0.5765)	0.9496 (0.0026)	0.9377 (0.0042)	0.0495 (0.0030)
suni2D_lambda_alpha	1.3611 (0.0344)	0.4381 (0.0114)	23.4050 (0.5769)	0.9488 (0.0026)	0.9387 (0.0042)	0.0504 (0.0030)
suni2D_sum_rate	1.3655 (0.0346)	0.4392 (0.0114)	23.5650 (0.5964)	0.9480 (0.0027)	0.9393 (0.0042)	0.0512 (0.0031)
suni2D_zeta_tau_std	1.3435 (0.0430)	0.5308 (0.0183)	32.7100 (0.9135)	0.9081 (0.0045)	0.9777 (0.0025)	0.0975 (0.0049)
suni2D_lambda_alpha_std	1.3437 (0.0432)	0.5299 (0.0182)	32.7800 (0.9625)	0.9079 (0.0048)	0.9783 (0.0025)	0.0979 (0.0052)
suni2D_sum_rate_std	1.3455 (0.0434)	0.5313 (0.0183)	32.9200 (0.9788)	0.9071 (0.0049)	0.9780 (0.0025)	0.0986 (0.0053)
uniLasso	1.3245 (0.0354)	0.4487 (0.0130)	24.5500 (0.5792)	0.9449 (0.0026)	0.9510 (0.0041)	0.0556 (0.0030)
uniLasso_loo	1.5295 (0.0366)	0.4255 (0.0119)	20.3950 (0.3866)	0.9628 (0.0016)	0.9317 (0.0043)	0.0347 (0.0019)
lasso	1.3392 (0.0422)	0.5251 (0.0171)	33.4300 (1.0169)	0.9043 (0.0050)	0.9763 (0.0025)	0.1015 (0.0055)
adalasso_uni	1.3782 (0.0343)	0.4354 (0.0113)	23.1900 (0.5880)	0.9489 (0.0026)	0.9323 (0.0042)	0.0498 (0.0030)
adalasso_lasso	1.4942 (0.0480)	0.4060 (0.0171)	16.5800 (0.3483)	0.9793 (0.0016)	0.9147 (0.0044)	0.0155 (0.0018)

Model 2

Huang, AR(1) rho = 0.95; n = 100, p = 200, 15 nonzero; sigma = 1.5

	sse	rpe	support	sign	tpr	fpr
suni2D_zeta_tau	5.7926 (0.1834)	0.1941 (0.0078)	15.5350 (0.2632)	0.9847 (0.0013)	0.9163 (0.0040)	0.0097 (0.0014)
suni2D_lambda_alpha	5.8088 (0.1846)	0.1962 (0.0079)	15.6300 (0.2672)	0.9842 (0.0013)	0.9163 (0.0040)	0.0102 (0.0014)
suni2D_sum_rate	5.7949 (0.1843)	0.1947 (0.0078)	15.5850 (0.2595)	0.9843 (0.0013)	0.9160 (0.0040)	0.0100 (0.0014)
suni2D_zeta_tau_std	6.0091 (0.2142)	0.2561 (0.0112)	19.1900 (0.5504)	0.9665 (0.0028)	0.9163 (0.0039)	0.0294 (0.0030)
suni2D_lambda_alpha_std	6.0168 (0.2159)	0.2554 (0.0112)	19.1750 (0.5533)	0.9666 (0.0028)	0.9163 (0.0039)	0.0294 (0.0030)
suni2D_sum_rate_std	6.0021 (0.2145)	0.2546 (0.0111)	19.2050 (0.5553)	0.9664 (0.0028)	0.9163 (0.0039)	0.0295 (0.0030)
uniLasso	5.8676 (0.1907)	0.2038 (0.0090)	15.6450 (0.2499)	0.9847 (0.0013)	0.9193 (0.0041)	0.0100 (0.0014)
uniLasso_loo	5.8899 (0.1907)	0.1884 (0.0073)	14.9600 (0.1697)	0.9881 (0.0009)	0.9197 (0.0040)	0.0063 (0.0009)
lasso	6.0855 (0.2166)	0.2624 (0.0120)	19.8450 (0.7725)	0.9638 (0.0039)	0.9200 (0.0038)	0.0327 (0.0042)
adalasso_uni	5.8830 (0.1910)	0.1938 (0.0079)	15.5500 (0.2540)	0.9852 (0.0013)	0.9213 (0.0038)	0.0094 (0.0013)
adalasso_lasso	9.8144 (0.3932)	0.3173 (0.0153)	14.3950 (0.3054)	0.9743 (0.0014)	0.8090 (0.0060)	0.0122 (0.0015)

Linear simulations with the tuned suni2D and uniLasso settings

Mean (standard error) across replicates. _std: suni2D with standardize = TRUE; _loo: uniLasso with LOO.

Model 3

Huang, 3 groups of 5 near−collinear signals; n = 100, p = 200; sigma = 1.5

	sse	rpe	support	sign	tpr	fpr
suni2D_zeta_tau	23.1024 (0.5669)	0.1902 (0.0059)	15.6500 (0.3610)	0.9697 (0.0018)	0.8193 (0.0058)	0.0182 (0.0019)
suni2D_lambda_alpha	23.1588 (0.5704)	0.1952 (0.0082)	15.8050 (0.4939)	0.9688 (0.0024)	0.8187 (0.0059)	0.0191 (0.0026)
suni2D_sum_rate	23.1931 (0.5662)	0.1952 (0.0082)	15.7400 (0.4953)	0.9689 (0.0024)	0.8177 (0.0058)	0.0188 (0.0026)
suni2D_zeta_tau_std	23.6666 (0.5605)	0.3694 (0.0120)	28.7700 (1.0488)	0.9023 (0.0052)	0.8080 (0.0060)	0.0900 (0.0056)
suni2D_lambda_alpha_std	23.6995 (0.5591)	0.3775 (0.0133)	29.2250 (1.1040)	0.9001 (0.0055)	0.8083 (0.0060)	0.0924 (0.0059)
suni2D_sum_rate_std	23.7048 (0.5616)	0.3688 (0.0119)	28.6000 (1.0403)	0.9030 (0.0052)	0.8063 (0.0060)	0.0892 (0.0056)
uniLasso	23.6772 (0.5771)	0.2170 (0.0080)	17.2700 (0.4728)	0.9622 (0.0022)	0.8240 (0.0060)	0.0265 (0.0024)
uniLasso_loo	26.1396 (0.6338)	0.2076 (0.0059)	15.2050 (0.2744)	0.9721 (0.0012)	0.8207 (0.0062)	0.0156 (0.0013)
lasso	24.1895 (0.5776)	0.3649 (0.0111)	28.9000 (1.0677)	0.9032 (0.0053)	0.8177 (0.0065)	0.0899 (0.0057)
adalasso_uni	23.9264 (0.5856)	0.1945 (0.0058)	15.7550 (0.3757)	0.9695 (0.0018)	0.8217 (0.0061)	0.0185 (0.0019)
adalasso_lasso	56.7092 (1.6229)	0.3550 (0.0123)	10.8850 (0.5476)	0.9479 (0.0023)	0.5160 (0.0083)	0.0170 (0.0026)

Model 4

uniLasso, equicorrelated rho = 0.5; n = 300, p = 1000, 30 nonzero ~ U(0, 1); sigma = 0.1

	sse	rpe	support	sign	tpr	fpr
suni2D_zeta_tau	0.0153 (0.0002)	0.8099 (0.0130)	132.7700 (1.4518)	0.8962 (0.0014)	0.9837 (0.0015)	0.1065 (0.0015)
suni2D_lambda_alpha	0.0146 (0.0002)	0.7685 (0.0129)	130.9300 (1.4435)	0.8981 (0.0014)	0.9835 (0.0015)	0.1046 (0.0015)
suni2D_sum_rate	0.0140 (0.0002)	0.7349 (0.0130)	125.7800 (1.3463)	0.9032 (0.0013)	0.9837 (0.0015)	0.0992 (0.0014)
suni2D_zeta_tau_std	0.0149 (0.0002)	0.7839 (0.0123)	129.9950 (1.3305)	0.8990 (0.0013)	0.9838 (0.0014)	0.1036 (0.0014)
suni2D_lambda_alpha_std	0.0143 (0.0002)	0.7542 (0.0121)	127.1750 (1.3676)	0.9019 (0.0014)	0.9840 (0.0015)	0.1007 (0.0014)
suni2D_sum_rate_std	0.0140 (0.0002)	0.7360 (0.0118)	124.4950 (1.1849)	0.9045 (0.0012)	0.9838 (0.0014)	0.0979 (0.0012)
uniLasso	0.0152 (0.0002)	0.8142 (0.0121)	124.2450 (0.6489)	0.9047 (0.0006)	0.9828 (0.0015)	0.0977 (0.0007)
uniLasso_loo	0.0140 (0.0002)	1.2189 (0.0169)	123.8000 (0.6414)	0.9052 (0.0006)	0.9828 (0.0015)	0.0972 (0.0007)
lasso	0.0155 (0.0002)	0.8325 (0.0128)	140.6650 (1.8621)	0.8883 (0.0018)	0.9835 (0.0015)	0.1146 (0.0019)
adalasso_uni	0.0157 (0.0002)	0.8527 (0.0129)	144.2800 (1.6764)	0.8847 (0.0017)	0.9833 (0.0015)	0.1183 (0.0017)
adalasso_lasso	0.0029 (0.0001)	0.1574 (0.0037)	32.1450 (0.2760)	0.9961 (0.0003)	0.9715 (0.0020)	0.0031 (0.0003)

Linear simulations with the tuned suni2D and uniLasso settings

Mean (standard error) across replicates. _std: suni2D with standardize = TRUE; _loo: uniLasso with LOO.

Model 5

uniLasso home court, AR(1) rho = 0.8; n = 100, p = 30, 20% nonzero; SNR = 1

	sse	rpe	support	sign	tpr	fpr
suni2D_zeta_tau	7.9934 (0.5502)	0.1414 (0.0050)	9.6800 (0.2411)	0.7647 (0.0074)	0.7208 (0.0126)	0.2231 (0.0091)
suni2D_lambda_alpha	8.0475 (0.5560)	0.1422 (0.0051)	9.7400 (0.2468)	0.7630 (0.0077)	0.7217 (0.0126)	0.2254 (0.0094)
suni2D_sum_rate	8.0531 (0.5561)	0.1424 (0.0051)	9.7550 (0.2477)	0.7628 (0.0077)	0.7225 (0.0126)	0.2258 (0.0094)
suni2D_zeta_tau_std	7.4323 (0.3871)	0.1449 (0.0056)	10.6900 (0.2639)	0.7405 (0.0081)	0.7442 (0.0128)	0.2594 (0.0101)
suni2D_lambda_alpha_std	7.4452 (0.3871)	0.1451 (0.0055)	10.6650 (0.2658)	0.7413 (0.0082)	0.7442 (0.0128)	0.2583 (0.0101)
suni2D_sum_rate_std	7.4441 (0.3869)	0.1451 (0.0055)	10.6550 (0.2660)	0.7417 (0.0082)	0.7442 (0.0128)	0.2579 (0.0102)
uniLasso	6.5507 (0.3238)	0.1315 (0.0045)	8.3600 (0.1609)	0.8015 (0.0058)	0.7008 (0.0131)	0.1731 (0.0061)
uniLasso_loo	6.3710 (0.3080)	0.1425 (0.0052)	6.9250 (0.1483)	0.8215 (0.0051)	0.6308 (0.0133)	0.1308 (0.0053)
lasso	7.6116 (0.4331)	0.1448 (0.0054)	10.8450 (0.2828)	0.7348 (0.0085)	0.7425 (0.0130)	0.2662 (0.0107)
adalasso_uni	8.3314 (0.5981)	0.1405 (0.0050)	9.8550 (0.2861)	0.7562 (0.0086)	0.7142 (0.0129)	0.2321 (0.0108)
adalasso_lasso	10.7692 (0.6723)	0.1845 (0.0079)	6.5950 (0.2700)	0.8075 (0.0068)	0.5692 (0.0151)	0.1325 (0.0092)

Model 6

Sign conflict: 3 blocks of 4 (rho = 0.5), (1, 1, 1, −1); n = 200, p = 500; SNR = 3

	sse	rpe	support	sign	tpr	fpr
suni2D_zeta_tau	3.0046 (0.0762)	0.4653 (0.0120)	34.9850 (1.0766)	0.9511 (0.0020)	0.9400 (0.0057)	0.0486 (0.0021)
suni2D_lambda_alpha	2.9961 (0.0761)	0.4644 (0.0120)	35.0150 (1.0789)	0.9511 (0.0020)	0.9404 (0.0056)	0.0486 (0.0021)
suni2D_sum_rate	2.9959 (0.0761)	0.4643 (0.0120)	35.0200 (1.0784)	0.9511 (0.0020)	0.9404 (0.0056)	0.0486 (0.0021)
suni2D_zeta_tau_std	4.2357 (0.0687)	0.6611 (0.0120)	57.2200 (1.9128)	0.9049 (0.0036)	0.9025 (0.0069)	0.0951 (0.0038)
suni2D_lambda_alpha_std	4.2312 (0.0689)	0.6608 (0.0120)	57.3100 (1.9259)	0.9047 (0.0036)	0.9029 (0.0069)	0.0952 (0.0038)
suni2D_sum_rate_std	4.2295 (0.0688)	0.6605 (0.0120)	57.2800 (1.9229)	0.9048 (0.0036)	0.9033 (0.0069)	0.0952 (0.0038)
uniLasso	5.7376 (0.0937)	0.9032 (0.0250)	50.1650 (2.0737)	0.9116 (0.0041)	0.7488 (0.0014)	0.0844 (0.0042)
uniLasso_loo	4.6992 (0.0337)	0.6508 (0.0089)	16.8150 (0.4588)	0.9782 (0.0009)	0.7462 (0.0012)	0.0161 (0.0009)
lasso	4.2240 (0.0684)	0.6605 (0.0120)	57.6350 (1.9361)	0.9042 (0.0036)	0.9067 (0.0068)	0.0958 (0.0038)
adalasso_uni	2.9942 (0.0760)	0.4641 (0.0119)	35.0700 (1.0753)	0.9511 (0.0020)	0.9417 (0.0055)	0.0487 (0.0021)
adalasso_lasso	3.3565 (0.1101)	0.4889 (0.0157)	17.2850 (0.5503)	0.9822 (0.0009)	0.8504 (0.0087)	0.0145 (0.0010)

Linear simulations with the tuned suni2D and uniLasso settings

Mean (standard error) across replicates. _std: suni2D with standardize = TRUE; _loo: uniLasso with LOO.

Model 7

Sign conflict: blocks of 10 (rho = 0.5), (1, 1, 1, −1) in 3 blocks; n = 400, p = 1000; SNR = 3

	sse	rpe	support	sign	tpr	fpr
suni2D_zeta_tau	1.0762 (0.0284)	0.1620 (0.0043)	38.3000 (0.8565)	0.9736 (0.0009)	0.9954 (0.0013)	0.0267 (0.0009)
suni2D_lambda_alpha	1.0695 (0.0282)	0.1613 (0.0043)	38.4600 (0.8667)	0.9734 (0.0009)	0.9954 (0.0013)	0.0268 (0.0009)
suni2D_sum_rate	1.0695 (0.0282)	0.1613 (0.0043)	38.4600 (0.8667)	0.9734 (0.0009)	0.9954 (0.0013)	0.0268 (0.0009)
suni2D_zeta_tau_std	1.7957 (0.0305)	0.2823 (0.0052)	84.0150 (1.5731)	0.9280 (0.0016)	1.0000 (0.0000)	0.0729 (0.0016)
suni2D_lambda_alpha_std	1.7938 (0.0305)	0.2822 (0.0052)	84.2050 (1.6005)	0.9278 (0.0016)	1.0000 (0.0000)	0.0731 (0.0016)
suni2D_sum_rate_std	1.7938 (0.0305)	0.2822 (0.0052)	84.2050 (1.6005)	0.9278 (0.0016)	1.0000 (0.0000)	0.0731 (0.0016)
uniLasso	4.5051 (0.0361)	0.6198 (0.0103)	44.0450 (1.9435)	0.9620 (0.0019)	0.7504 (0.0004)	0.0355 (0.0020)
uniLasso_loo	4.1231 (0.0191)	0.5372 (0.0058)	16.7400 (0.5145)	0.9893 (0.0005)	0.7500 (0.0000)	0.0078 (0.0005)
lasso	1.7920 (0.0302)	0.2817 (0.0052)	83.9550 (1.5480)	0.9280 (0.0015)	1.0000 (0.0000)	0.0728 (0.0016)
adalasso_uni	1.0746 (0.0284)	0.1617 (0.0043)	38.0850 (0.8586)	0.9738 (0.0009)	0.9954 (0.0013)	0.0265 (0.0009)
adalasso_lasso	0.5197 (0.0182)	0.0834 (0.0030)	16.9450 (0.2807)	0.9950 (0.0003)	0.9992 (0.0006)	0.0050 (0.0003)

Model 8

Sign conflict: blocks of 10 (rho = 0.7), (1, 1, 1, −1.5) in 3 blocks; n = 400, p = 1000; SNR = 3

	sse	rpe	support	sign	tpr	fpr
suni2D_zeta_tau	1.1330 (0.0254)	0.1421 (0.0034)	35.9700 (0.7815)	0.9760 (0.0008)	1.0000 (0.0000)	0.0243 (0.0008)
suni2D_lambda_alpha	1.1297 (0.0253)	0.1418 (0.0034)	35.9650 (0.7935)	0.9760 (0.0008)	1.0000 (0.0000)	0.0243 (0.0008)
suni2D_sum_rate	1.1297 (0.0253)	0.1418 (0.0034)	35.9650 (0.7935)	0.9760 (0.0008)	1.0000 (0.0000)	0.0243 (0.0008)
suni2D_zeta_tau_std	2.4740 (0.0441)	0.3368 (0.0058)	98.7650 (1.2584)	0.9132 (0.0013)	1.0000 (0.0000)	0.0878 (0.0013)
suni2D_lambda_alpha_std	2.4713 (0.0441)	0.3364 (0.0058)	98.7600 (1.2574)	0.9132 (0.0013)	1.0000 (0.0000)	0.0878 (0.0013)
suni2D_sum_rate_std	2.4713 (0.0441)	0.3364 (0.0058)	98.7600 (1.2574)	0.9132 (0.0013)	1.0000 (0.0000)	0.0878 (0.0013)
uniLasso	9.5135 (0.0334)	0.9863 (0.0135)	37.0650 (1.4764)	0.9689 (0.0015)	0.7467 (0.0013)	0.0284 (0.0015)
uniLasso_loo	9.2479 (0.0261)	0.9068 (0.0090)	15.1000 (0.4064)	0.9908 (0.0004)	0.7442 (0.0016)	0.0062 (0.0004)
lasso	2.4693 (0.0440)	0.3364 (0.0058)	98.9050 (1.2602)	0.9131 (0.0013)	1.0000 (0.0000)	0.0880 (0.0013)
adalasso_uni	1.1287 (0.0253)	0.1418 (0.0035)	36.1150 (0.7957)	0.9759 (0.0008)	1.0000 (0.0000)	0.0244 (0.0008)
adalasso_lasso	0.6088 (0.0211)	0.0792 (0.0025)	17.1000 (0.2927)	0.9949 (0.0003)	0.9996 (0.0004)	0.0052 (0.0003)

Linear simulations with the tuned suni2D and uniLasso settings

Mean (standard error) across replicates. _std: suni2D with standardize = TRUE; _loo: uniLasso with LOO.

Model 9

Sign conflict: blocks of 10 (rho = 0.8), (2, 2, −2) in 3 blocks; n = 400, p = 1000; SNR = 3

	sse	rpe	support	sign	tpr	fpr
suni2D_zeta_tau	2.0004 (0.0510)	0.0999 (0.0025)	26.6000 (0.5241)	0.9824 (0.0005)	1.0000 (0.0000)	0.0178 (0.0005)
suni2D_lambda_alpha	1.9971 (0.0511)	0.0997 (0.0025)	26.6600 (0.5392)	0.9823 (0.0005)	1.0000 (0.0000)	0.0178 (0.0005)
suni2D_sum_rate	1.9971 (0.0511)	0.0997 (0.0025)	26.6600 (0.5392)	0.9823 (0.0005)	1.0000 (0.0000)	0.0178 (0.0005)
suni2D_zeta_tau_std	5.7971 (0.1119)	0.3357 (0.0061)	96.8350 (1.2861)	0.9122 (0.0013)	1.0000 (0.0000)	0.0886 (0.0013)
suni2D_lambda_alpha_std	5.7881 (0.1118)	0.3353 (0.0061)	96.8300 (1.2876)	0.9122 (0.0013)	1.0000 (0.0000)	0.0886 (0.0013)
suni2D_sum_rate_std	5.7881 (0.1118)	0.3353 (0.0061)	96.8300 (1.2876)	0.9122 (0.0013)	1.0000 (0.0000)	0.0886 (0.0013)
uniLasso	18.2308 (0.0533)	0.7381 (0.0095)	26.8700 (1.2017)	0.9761 (0.0012)	0.6667 (0.0000)	0.0211 (0.0012)
uniLasso_loo	17.8365 (0.0445)	0.6780 (0.0076)	10.6250 (0.3528)	0.9924 (0.0004)	0.6667 (0.0000)	0.0047 (0.0004)
lasso	5.7835 (0.1112)	0.3349 (0.0060)	97.1600 (1.2837)	0.9118 (0.0013)	1.0000 (0.0000)	0.0890 (0.0013)
adalasso_uni	2.0058 (0.0515)	0.0998 (0.0025)	26.4000 (0.5310)	0.9826 (0.0005)	1.0000 (0.0000)	0.0176 (0.0005)
adalasso_lasso	1.2222 (0.0522)	0.0656 (0.0025)	13.6700 (0.2649)	0.9953 (0.0003)	1.0000 (0.0000)	0.0047 (0.0003)

Model 10

Sign conflict: AR(1) rho = 0.8, (1, 1, 1, −1.5) at 3 places; n = 400, p = 1000; SNR = 3

	sse	rpe	support	sign	tpr	fpr
suni2D_zeta_tau	2.7712 (0.0944)	0.2170 (0.0061)	46.3650 (1.1374)	0.9651 (0.0011)	0.9771 (0.0038)	0.0351 (0.0011)
suni2D_lambda_alpha	2.7408 (0.0914)	0.2152 (0.0060)	46.5000 (1.1394)	0.9650 (0.0011)	0.9792 (0.0036)	0.0352 (0.0011)
suni2D_sum_rate	2.7408 (0.0914)	0.2152 (0.0060)	46.5000 (1.1394)	0.9650 (0.0011)	0.9792 (0.0036)	0.0352 (0.0011)
suni2D_zeta_tau_std	4.6649 (0.0931)	0.3631 (0.0058)	93.6400 (1.8433)	0.9172 (0.0018)	0.9504 (0.0046)	0.0832 (0.0018)
suni2D_lambda_alpha_std	4.6613 (0.0930)	0.3626 (0.0058)	93.5600 (1.8403)	0.9173 (0.0018)	0.9504 (0.0046)	0.0832 (0.0018)
suni2D_sum_rate_std	4.6613 (0.0930)	0.3626 (0.0058)	93.5600 (1.8403)	0.9173 (0.0018)	0.9504 (0.0046)	0.0832 (0.0018)
uniLasso	10.5476 (0.0333)	0.7800 (0.0101)	36.6900 (1.8312)	0.9640 (0.0018)	0.5304 (0.0037)	0.0307 (0.0018)
uniLasso_loo	10.3109 (0.0230)	0.6995 (0.0074)	12.1150 (0.4243)	0.9882 (0.0004)	0.5142 (0.0024)	0.0060 (0.0004)
lasso	4.6590 (0.0930)	0.3627 (0.0058)	93.9550 (1.8373)	0.9169 (0.0018)	0.9513 (0.0046)	0.0835 (0.0018)
adalasso_uni	2.7482 (0.0912)	0.2157 (0.0060)	46.6350 (1.1679)	0.9648 (0.0012)	0.9779 (0.0037)	0.0353 (0.0012)
adalasso_lasso	2.9544 (0.1073)	0.1586 (0.0048)	17.4500 (0.3611)	0.9923 (0.0003)	0.9071 (0.0060)	0.0066 (0.0003)